CSE 270 Sports Analytics

Measuring individual performance in team sports

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NBA Individual Power Ratings

NBA Individual Power Ratings

Libraries used

```
library(SportsAnalytics270)
library(dplyr)
library(ggplot2)
library(VGAM)
library(zoo)
library(tidyr)
# remotes::install_github("Habet/CSE270")
```

NBA Individual Power Ratings

2 main stages of the game.

- Attacking
- Defending

https://www.youtube.com/watch?v=wYjp2zoqQrs

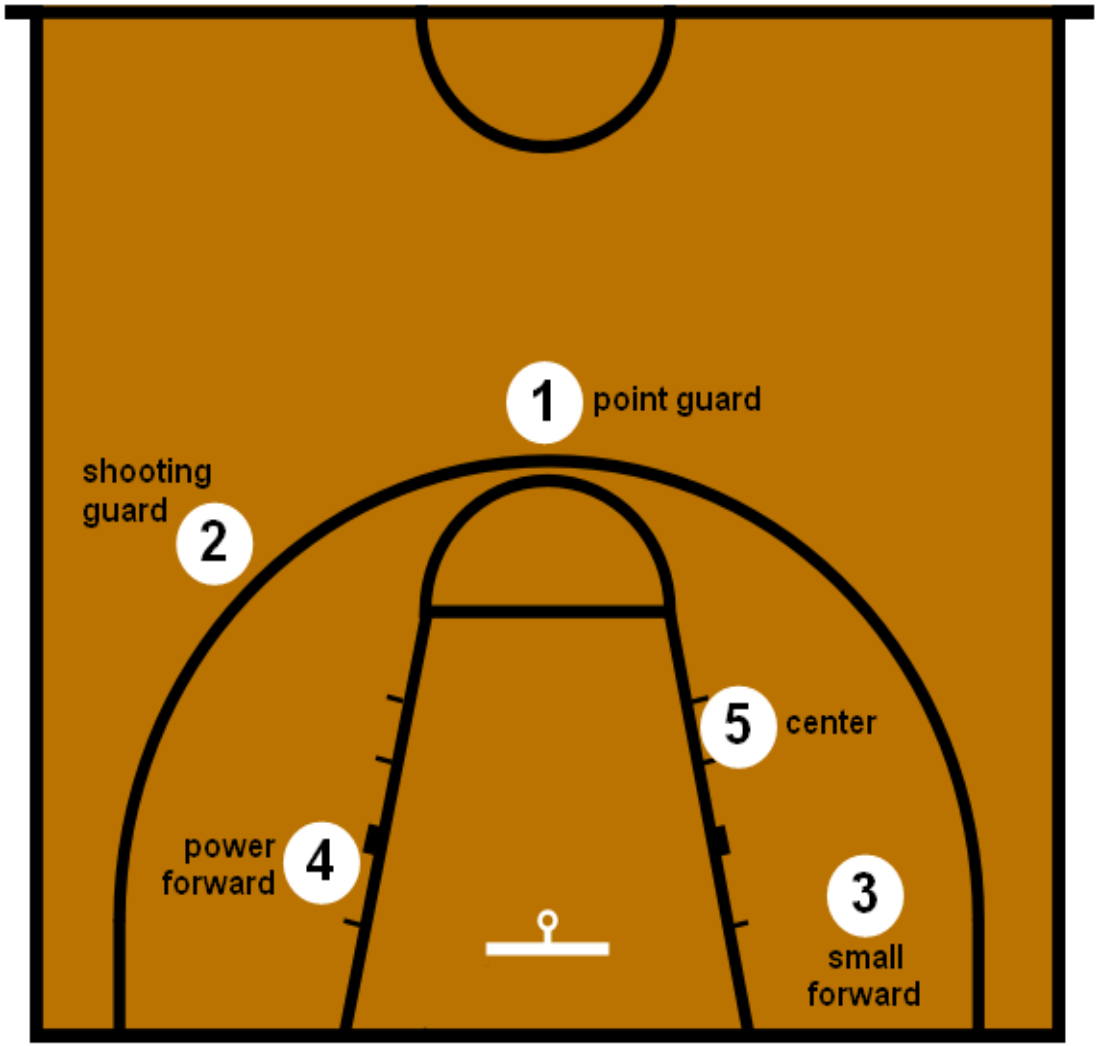
NBA Individual Power Ratings

Five positions. Vote on which box-score column belongs to each one before the next slide.

- Center: close shots, rebounds, blocks
- Power forward: the same area, with longer shots
- Small forward: both ends of the floor
- Point guard: runs the offense, defends the ball, steals
- Shooting guard: long shots and the dribble

Short video: https://www.youtube.com/watch?v=4_4CymXARWQ

NBA Individual Power Ratings



NBA Individual Power Ratings

What, other than height, would you use to place a player?

NBA Individual Power Ratings

- Center and power forward: blocks and defensive rebounds
- Point guard: assists and steals
- Wings: three-point attempts and makes

Guess first. The box score is the check.

NBA Individual Power Ratings

In game statistics

- STEALS
- TURNOVERS - Turnover occurs when a team loses possession of the ball to the opposing team before a player takes a shot at their team's basket.
  https://www.youtube.com/shorts/I4MDXOWrUmk
- REBOUNDS
  1. OFFENSIVE REBOUNDS - An offensive rebound occurs when offensive players regain possession of the ball after a missed shot.
  2. DEFENSIVE REBOUNDS - Defensive rebounds occur when defensive players gain possession of the ball after an offensive player misses the shot.
- BLOCKS - Block occurs when a defensive player legally deflects a field goal attempt from an offensive player to prevent a score.
- FOULS
  1. PERSONAL FOULS
  2. FLAGRANT FOULS
  3. TECHINCAL FOULS

NBA Individual Power Ratings

Dataset: `nba_players`

Season box scores from Basketball-Reference, through 2024-25. One row is one player-season. `Year` is the season ending year, so 2025 is 2024-25.

- Identity: `Year`, `Player`, `Pos`, `Tm`
- Time: `G` games, `GS` starts, `MP` minutes
- Defense: `STL`, `TOV`, `DRB`, `BLK`, `PF`
- Offense: `PTS`, `AST`, `ORB`, `FG/FGA`, `X3P/X3PA`, `FT/FTA`
- Rates end with a dot: `FG.` is field-goal percentage

NBA Individual Power Ratings

Counting totals is not a rating.

- A total grows when a player stays on the floor
- Points and minutes move together for that reason
- Dividing by minutes asks what he did per minute

The next slides use one season of totals on purpose, so the trap is visible. They are not a five-year average, and they are not a success rate.

NBA Individual Power Ratings

Let's start with defending attributes.

Successful actions, when defending

- STEALS
- BLOCKS
- DEFENSIVE REBOUNDS

Non Successful actions, when defending

- PERSONAL FOULS
- TURNOVER

NBA Individual Power Ratings

Let's calculate the difference between the cumulative success actions and cumulative failure actions.

```
nba_players_defence <- nba_players %>%
  arrange(Player, Year) %>%
  filter(Pos %in% c('SF','PG')) %>%
  group_by(Player) %>%
  mutate(STL_TOV = STL - TOV,
         STL_PF = STL - PF,
         DRB_TOV = DRB - TOV,
         BLK_TOV = BLK - TOV,
         BLK_PF = BLK - PF) %>%
  drop_na(STL_TOV, STL_PF, DRB_TOV, BLK_TOV, BLK_PF, MP) %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_defence$MP))
}

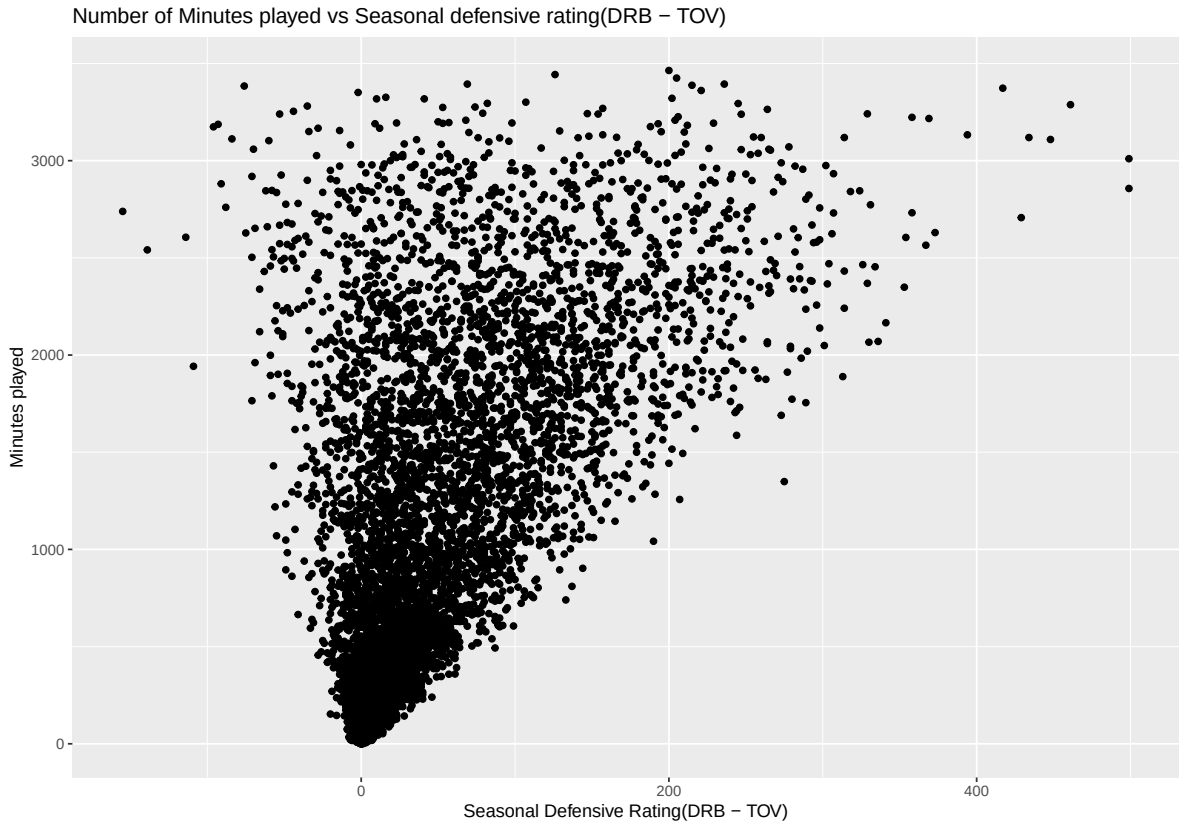
nba_players_defence %>%
  select(STL_TOV, STL_PF, DRB_TOV, BLK_TOV, BLK_PF) %>%
  apply(2,cor_with_MP)
```

```
##    STL_TOV     STL_PF    DRB_TOV    BLK_TOV     BLK_PF
## -0.6490088 -0.6950625  0.5825263 -0.8132951 -0.8773959
```

NBA Individual Power Ratings

The metric with the highest positive correlation rate is the Seasonal difference between the seasonal number of defensive rebounds made and number of turnovers made.

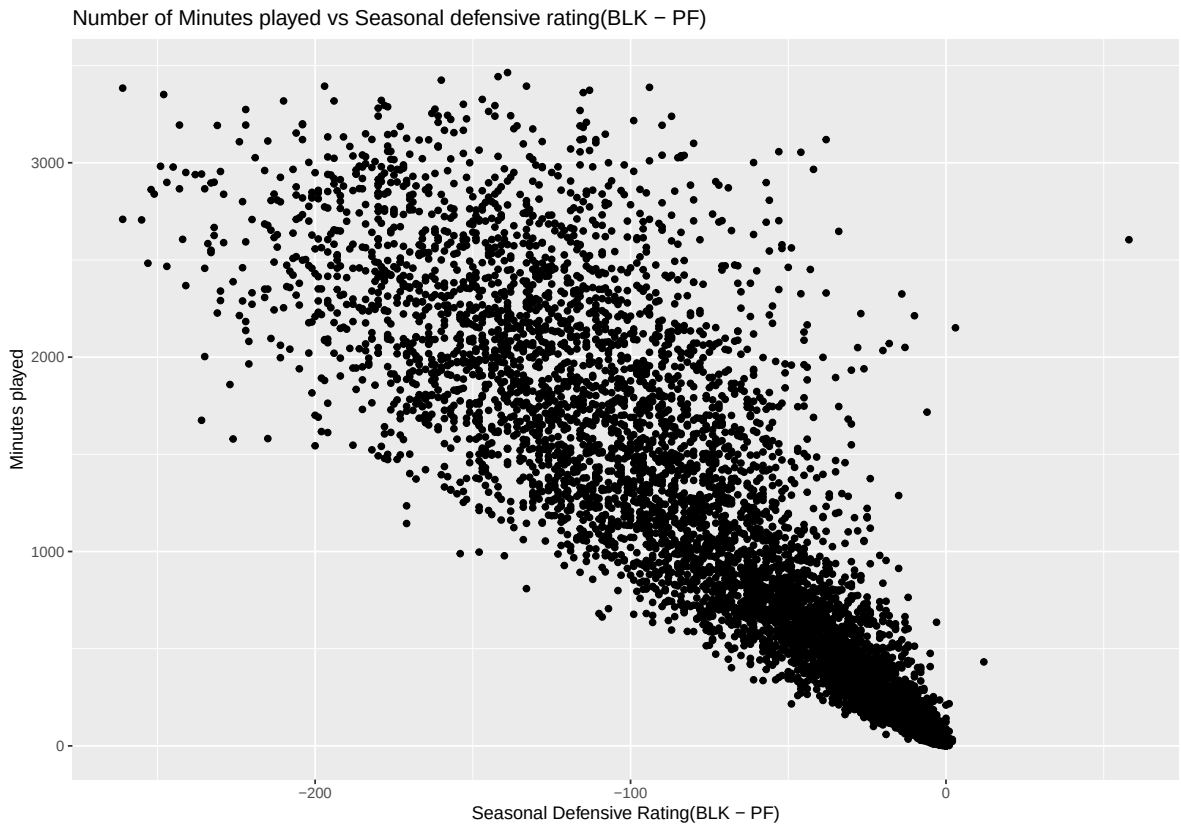
```
ggplot(nba_players_defence, aes(x = DRB_TOV, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(DRB - TOV)") +
  xlab("Seasonal Defensive Rating(DRB - TOV)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

The metric with the highest negative correlation is blocks minus personal fouls.

```
ggplot(nba_players_defence, aes(x = BLK_PF, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(BLK - PF)") +
  xlab("Seasonal Defensive Rating(BLK - PF)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

Let's fit a model with the defensive power ratings.

```
summary(lm(MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF ), data = nba_players_defence))
```

```
##
## Call:
## lm(formula = MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF), data = nba_players_defence)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2405.43  -111.51   -56.29    91.06  2330.39
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  74.40429    5.42980   13.70   <2e-16 ***
## STL_TOV      -6.75886    0.10654  -63.44   <2e-16 ***
## STL_PF        8.30940    0.17114   48.55   <2e-16 ***
## DRB_TOV       4.37224    0.05939   73.61   <2e-16 ***
## BLK_PF      -13.91125    0.12813 -108.57   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 273.5 on 6516 degrees of freedom
## Multiple R-squared:  0.9045, Adjusted R-squared:  0.9045
## F-statistic: 1.543e+04 on 4 and 6516 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

Read the defense regression as playing time, not as a cause.

- Season totals grow with minutes, so the correlations are expected
- More rebounds do not, by themselves, “earn” more minutes
- A rating has to divide by minutes before it ranks players

NBA Individual Power Ratings

For the attacking attributes we will calculate the difference between the attempts made for successful actions and actual successful actions.

Successful actions, when attacking

- FIELD GOALS
- 3 POINT GOALS
- FREE THROWS
- OFFENSIVE REBOUNDS
- POINTS SCORED
- ASSISTS MADE

Non Successful actions, when attacking

- FIELD GOALS ATTEMPTS
- 3 POINT ATTEMPTS
- FREE THROW ATTEMPTS

NBA Individual Power Ratings

```
nba_players_offense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(FGA_FG = FGA - FG,
         X3PA_X3P = X3PA - X3P,
         FTA_FT = FTA - FT) %>%
  drop_na(FGA_FG, X3PA_X3P, FTA_FT, AST, PTS, MP) %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the Seasonal metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_offense$MP))
}

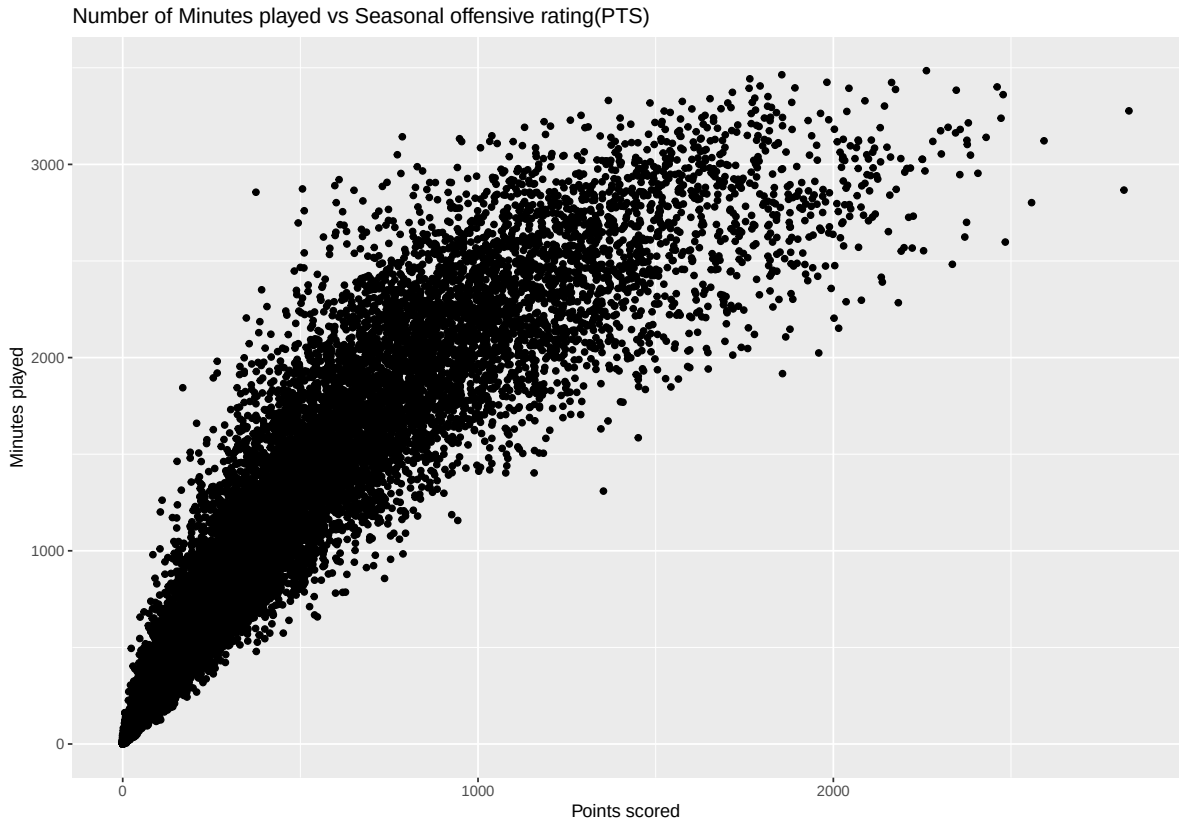
nba_players_offense %>%
  select(FGA_FG, X3PA_X3P, FTA_FT, AST, PTS) %>%
  apply(2,cor_with_MP)
```

```
##    FGA_FG  X3PA_X3P    FTA_FT       AST       PTS
## 0.9233937 0.6532211 0.7078765 0.7545190 0.9233347
```

NBA Individual Power Ratings

Missed field goals (FGA - FG) and points scored have the strongest correlations with minutes, both about 0.92. The plot shows points.

```
ggplot(nba_players_offense,aes(x = PTS, y = MP)) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal offensive rating(PTS)") +
  xlab("Points scored") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

2024-25 scoring list. Second in points: is he second per minute?

Vote before the tables. Anthony Edwards plays the most minutes of this group.

Points, players with more than 500 minutes

Player	Pos	Tm	MP	PTS	PTS36
Shai Gilgeous-Alexander	PG	OKC	2598	2484	34.4
Anthony Edwards	SG	MIN	2871	2177	27.3
Nikola Jokić	C	DEN	2571	2071	29.0
Giannis Antetokounmpo	PF	MIL	2289	2036	32.0
Jayson Tatum	PF	BOS	2624	1932	26.5

NBA Individual Power Ratings

Points per 36 minutes, players with more than 1500 minutes.

Player	Pos	Tm	MP	PTS	PTS36
Shai Gilgeous-Alexander	PG	OKC	2598	2484	34.4
Giannis Antetokounmpo	PF	MIL	2289	2036	32.0
Nikola Jokić	C	DEN	2571	2071	29.0
Luka Dončić	PG	2TM	1769	1408	28.7
LaMelo Ball	PG	CHO	1505	1184	28.3

- Shai Gilgeous-Alexander leads both lists
- Edwards is second in points and off this list
- Same season, same players: the denominator changes the ranking

NBA Individual Power Ratings

```
summary(lm(MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2307.83  -189.34   -67.36   156.27  1635.69
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 215.99824    3.40124  63.506  < 2e-16 ***
## PTS           0.47187    0.02557  18.453  < 2e-16 ***
## AST           0.68208    0.03050  22.365  < 2e-16 ***
## FGA_FG        2.44019    0.05819  41.932  < 2e-16 ***
## X3PA_X3P     -0.16328    0.05241  -3.115  0.00184 **
## FTA_FT        3.46688    0.13691  25.323  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 304.4 on 16941 degrees of freedom
## Multiple R-squared:  0.8754, Adjusted R-squared:  0.8754
## F-statistic: 2.38e+04 on 5 and 16941 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

The offense regression has the same trap.

- Scoring more is not what puts a player on the floor
- Minutes produce points, assists, and missed shots
- Use the per-36 table, not this coefficient, when you rank scorers

Plus/Minus

Plus/Minus

The concept

Plus-minus measures the point differential for the team when the player is on the court,
Point differential is the difference in the points scored for and the points scored against the team.
The Montreal Canadiens tracked plus/minus in the 1950s. The NHL made it official for 1967-68.
A goal total ignores the goals against while that player was on the ice.

One stint, by hand

Home scores 2. Away scores 0. Durant is at home.

- Each home player, including Durant: +2
- Each away player: -2
- A player who is sitting: 0 for this stint

Season plus/minus is the sum of the stints he actually played.

Plus/minus for NBA

This file is the 2011-12 lockout season, from 25 December 2011 to 26 April 2012.

- One row is a stint: the same ten players stay on the floor
- Coverage is uneven. New Jersey has 28 games in the file, Memphis has 66
- The Finals are not in the file. That June, Miami beat Oklahoma City
- The number you compute is for these stints, not the official season total

The stint file: `nba12`

25,300 rows. One row is one stretch of 2011-12 with the same ten players on the floor.

```
data(nba12, package = "SportsAnalytics270")
```

- Game: `GameID`, `Date`, `AwayT`, `HomeT`
- Who is on the floor: `HomePlayer1Name` to `HomePlayer5Name`, same for away, and matching `...ID` columns
- Scoring: `PointsScoredHome`, `PointsScoredAway`
- Length: `ElapsedSecs`, `PossessionsHome`, `PossessionsAway`

Three stints, Christmas, Knicks at home

Boston is away. `PossessionsHome` counts New York trips; `PossessionsAway` counts Boston trips.

HomeT	AwayT	PointsScoredHome	PointsScoredAway	PossessionsHome	PossessionsAway
NYK	BOS	0	2	0	1
NYK	BOS	0	0	1	0
NYK	BOS	2	0	1	1

- Row 1. `PossessionsHome = 0`: New York did not have the ball. Boston did, and scored 2.
- Row 2. `PossessionsHome = 1`: New York had the ball once. Boston had 0 possessions.
- Row 3. Both equal 1: each team had one trip. New York scored 2, Boston scored 0.

The player file: `nba12p`

One row is one player in that stint file. Use it to attach names and minutes to player IDs.

```
data(nba12p, package = "SportsAnalytics270")
```

- `PlayerID` matches the ID columns in `nba12`
- `PlayerTrueName`, `PlayerTeams`
- `SimpleMin`: minutes in these stints, not necessarily the official season total

Four players from `nba12p`

`PlayerID` 714 is Durant. That is the number in the `nba12` lineup columns.

```
nba12p %>%
  filter(PlayerTrueName %in% c("Durant, Kevin", "James, LeBron",
                               "Westbrook, Russell", "Benson, Keith")) %>%
  select(PlayerID, PlayerTrueName, PlayerTeams, SimpleMin)
```

```
##   PlayerID     PlayerTrueName PlayerTeams SimpleMin
## 1     1027      Benson, Keith         GSW    9.1333
## 2      260      James, LeBron         MIA 2190.1000
## 3      786 Westbrook, Russell         OKC 2234.2000
## 4      714      Durant, Kevin         OKC 2430.9000
```

Oklahoma City from `nba12`

`GameID` is the date plus the two team codes, for example `201112250RLOKC`. Keep the rows that contain `OKC`.

```
okc <- nba12 %>%
  filter(grepl("OKC", GameID, fixed = TRUE))
nrow(okc)
```

```
## [1] 1716
```

```
okc %>%
  select(GameID, Date, HomeT, AwayT) %>%
  slice(1:3)
```

```
##           GameID     Date HomeT AwayT
## 1 201112250RLOKC 20111225   OKC   ORL
## 2 201112250RLOKC 20111225   OKC   ORL
## 3 201112250RLOKC 20111225   OKC   ORL
```

1,716 stints, from Thunder games only.

Stint plus/minus

Same scoring columns as in `nba12`. Home margin is points scored minus points allowed. Away margin is the opposite.

```
okc <- okc %>%
  mutate(
    Home_PM = PointsScoredHome - PointsScoredAway,
    Away_PM = -Home_PM
  )
okc %>%
  filter(PointsScoredHome + PointsScoredAway > 0) %>%
  select(HomeT, AwayT, PointsScoredHome, PointsScoredAway, Home_PM, Away_PM) %>%
  slice(1:3)
```

```
##   HomeT AwayT PointsScoredHome PointsScoredAway Home_PM Away_PM
## 1   OKC   ORL               18               18       0       0
## 2   OKC   ORL                0                3      -3       3
## 3   OKC   ORL                2                0       2      -2
```

First scoring stint: OKC at home, 18-18, so both margins are 0. A later 2-0 stint would be `Home_PM = 2`.

Is Durant on the floor?

Paste all five names, then search the string. Still the OKC rows, not the full `nba12` file.

```
okc <- okc %>%
  mutate(
    line_h = paste(HomePlayer1Name, HomePlayer2Name, HomePlayer3Name,
                   HomePlayer4Name, HomePlayer5Name, sep = ","),
    line_a = paste(AwayPlayer1Name, AwayPlayer2Name, AwayPlayer3Name,
                   AwayPlayer4Name, AwayPlayer5Name, sep = ","),
    D_H = grepl("Durant", line_h, fixed = TRUE),
    D_A = grepl("Durant", line_a, fixed = TRUE)
  )
```

Is Durant on the floor?

Christmas, one stint. These are the five names that went into `line_h` and `line_a`.

Home	H1	H2	H3	H4	H5	D_H
OKC	Westbrook	Sefolosha	Durant	Ibaka	Perkins	TRUE

Away	A1	A2	A3	A4	A5	D_A
ORL	Nelson	J. Richardson	Turkoglu	Anderson	Howard	FALSE

- `D_H = TRUE`: H3 is Durant, so `"Durant"` is in `line_h`
- `D_A = FALSE`: Durant is not in Orlando's five, so this stint uses `Home_PM`

Plus/minus for NBA

Durant's season plus/minus in this file: add the stint margin when he is on the floor.

```
# D_H / D_A: Durant is in the home or away five for that stint
durant_season <- sum(okc$Home_PM[okc$D_H]) + sum(okc$Away_PM[okc$D_A])
durant_season
```

```
## [1] 324
```

This is his plus/minus over the stints in this file, not the official 2011-12 total.

Plus/minus for NBA

LeBron needs a player id. Three players in this file have the last name James.

```
# nba12p has three players named James; match the full name
lebron_id <- nba12p$PlayerID[nba12p$PlayerTrueName == "James, LeBron"]
lebron_season <- player_pm(nba12, lebron_id)
c(Durant = durant_season, LeBron = lebron_season)
```

```
## Durant LeBron
##    324    414
```

Plus/minus for NBA

One number per Oklahoma City game, in date order.

```
durant_pm <- okc %>%
  # this stint: home margin if Durant is home, away margin if he is away
  mutate(PM = ifelse(D_H, Home_PM, ifelse(D_A, Away_PM, 0))) %>%
  group_by(GameID) %>%
  summarise(
    PM = sum(PM),  # add the stints in that game
    Date = as.Date(as.character(min(Date)), format = "%Y%m%d"),
    .groups = "drop"
  ) %>%
  arrange(Date) %>%
  mutate(Game = row_number())  # 1, 2, 3... for the bar chart
```

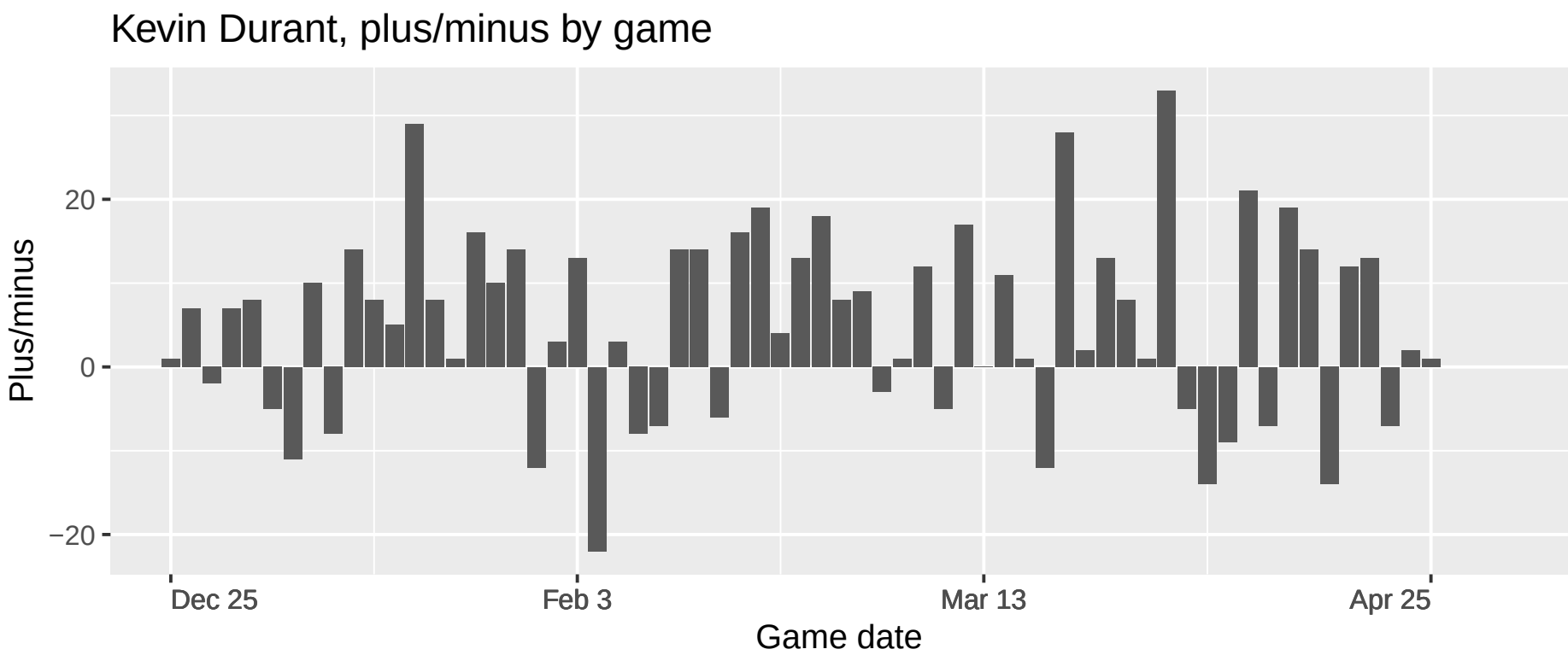
Plus/minus for NBA

```
head(durant_pm, n=5)
```

```
## # A tibble: 5 x 4
##   GameID            PM Date        Game
##   <fct>          <dbl> <date>     <int>
## 1 20111225ORLOKC     1 2011-12-25     1
## 2 20111226OKCMIN     7 2011-12-26     2
## 3 20111228OKCMEM    -2 2011-12-28     3
## 4 20111229DALOKC     7 2011-12-29     4
## 5 20111231PHXOKC     8 2011-12-31     5
```

Plus/minus for NBA

Each bar is one Oklahoma City game in this file, in date order.



On court and off court +/- and Roland's rating

The on court +/- number represents the team's net points with the player on the floor per 48 minutes, while the off court number is the team's net with the player off the floor per 48 minutes.

The Roland Rating is the difference between the two, with a positive number indicating the team has played better with the player than without.

OKC's margin, and whether Durant played

OKC is home in some rows and away in others. `OKC_PM` is always Thunder points minus opponent points. `D_PM` is TRUE if Durant was in either five.

```
okc <- okc %>%
  mutate(
    OKC_PM = ifelse(AwayT == "OKC", Away_PM,
                    ifelse(HomeT == "OKC", Home_PM, 0)),
    D_PM = D_H | D_A
  )
okc %>%
  filter(OKC_PM != 0) %>%
  group_by(HomeT) %>%
  slice(1) %>%
  ungroup() %>%
  select(HomeT, AwayT, D_H, D_PM, Home_PM, Away_PM, OKC_PM)
```

```
## # A tibble: 23 x 7
##    HomeT AwayT D_H   D_PM  Home_PM Away_PM OKC_PM
##    <chr> <chr> <lgl> <lgl>   <int>   <int>  <dbl>
##  1 ATL   OKC   FALSE TRUE        3      -3     -3
##  2 BOS   OKC   FALSE TRUE       -1       1      1
##  3 DAL   OKC   FALSE TRUE        1      -1     -1
##  4 DEN   OKC   FALSE TRUE       -6       6      6
##  5 GSW   OKC   FALSE TRUE       -4       4      4
##  6 HOU   OKC   FALSE TRUE       -4       4      4
##  7 IND   OKC   FALSE TRUE        5      -5     -5
##  8 LAC   OKC   FALSE TRUE        7      -7     -7
##  9 LAL   OKC   FALSE TRUE        6      -6     -6
## 10 MEM   OKC   FALSE TRUE       -3       3      3
## # i 13 more rows
```

When OKC is home, `OKC_PM` equals `Home_PM`. When OKC is away, it equals `Away_PM`.

Net Plus/minus for NBA

First half: one row for Durant on the floor, one row for Durant on the bench.

```
durant_long <- okc %>%
  group_by(GameID, D_PM) %>%
  summarise(
    PM = sum(OKC_PM),  # OKC margin while he is on, then while he is off
    Date = as.Date(as.character(min(Date)), format = "%Y%m%d"),
    .groups = "drop"
  ) %>%
  mutate(court = ifelse(D_PM, "on", "off")) %>%
  select(GameID, Date, court, PM)
head(durant_long, 4)
```

```
## # A tibble: 4 x 4
##   GameID         Date       court    PM
##   <fct>          <date>     <chr> <dbl>
## 1 201112250RLOKC 2011-12-25 off       7
## 2 201112250RLOKC 2011-12-25 on        1
## 3 201112260KCMIN 2011-12-26 off      -3
## 4 201112260KCMIN 2011-12-26 on        7
```

Christmas is two rows: off $+7$, on $+1$. Not yet subtracted.

What `pivot_wider` does

A toy game with the same two rows.

```
toy <- data.frame(
  Game = c("Christmas", "Christmas"),
  court = c("on", "off"),
  PM = c(1, 7)
)
toy
```

```
##        Game court PM
## 1 Christmas    on  1
## 2 Christmas   off  7
```

```
tidyr::pivot_wider(toy, names_from = court, values_from = PM)
```

```
## # A tibble: 1 x 3
##   Game          on   off
##   <chr>      <dbl> <dbl>
## 1 Christmas      1     7
```

`names_from = court` turns the words `"on"` and `"off"` into column names. `values_from = PM` fills those columns. Two rows become one.

Net Plus/minus for NBA

Second half: widen, then subtract. A game with no bench stint has `off = NA` and drops out.

```
durant_net <- durant_long %>%
  tidyr::pivot_wider(names_from = court, values_from = PM) %>%
  mutate(NET = on - off) %>%  # Roland's rating for that game
  filter(is.finite(NET)) %>%
  arrange(Date) %>%
  mutate(Game = row_number())
```

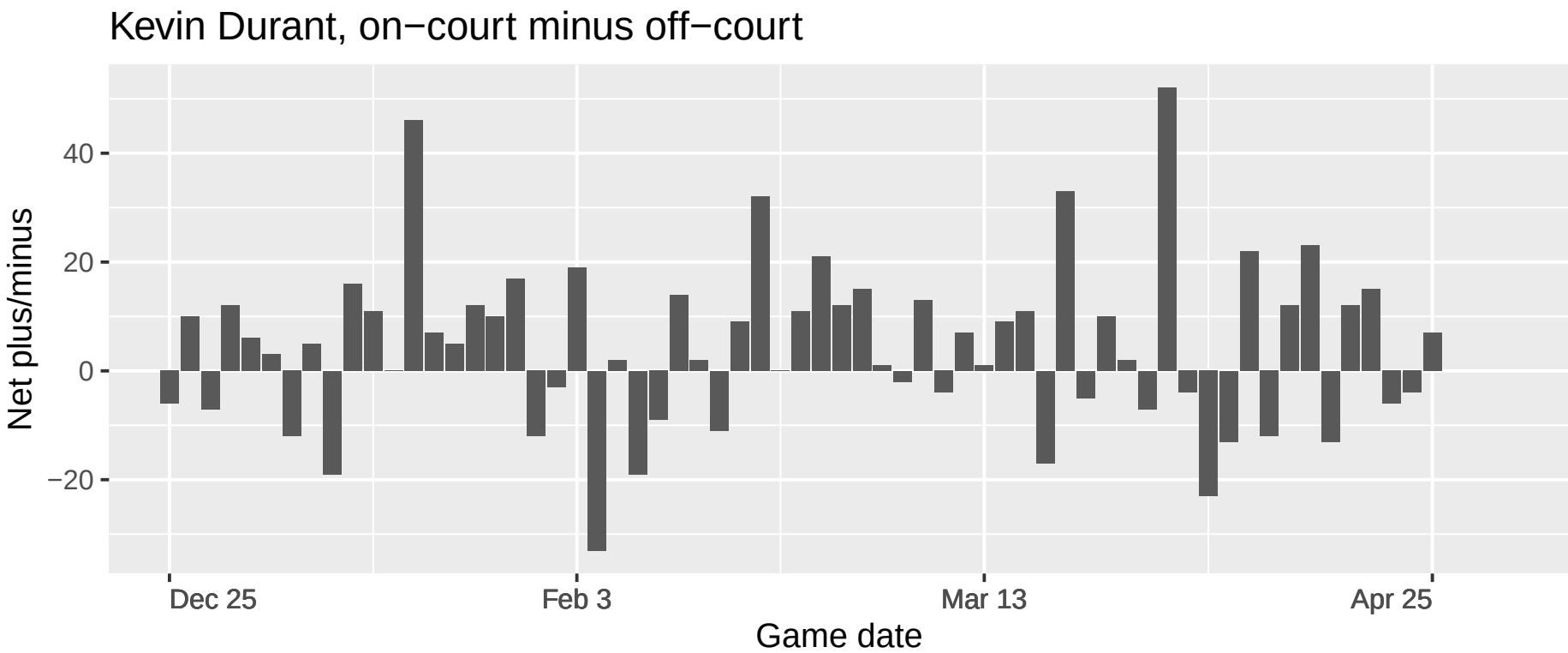
Net Plus/minus for NBA

Christmas: OKC was +1 with Durant on the floor and +7 with him off, so NET is −6.

```
head(durant_net)
```

```
## # A tibble: 6 x 6
##   GameID         Date         off    on   NET  Game
##   <fct>          <date>     <dbl> <dbl> <dbl> <int>
## 1 20111225ORLOKC 2011-12-25     7     1    -6     1
## 2 20111226OKCMIN 2011-12-26    -3     7    10     2
## 3 20111228OKCMEM 2011-12-28     5    -2    -7     3
## 4 20111229DALOKC 2011-12-29    -5     7    12     4
## 5 20111231PHXOKC 2011-12-31     2     8     6     5
## 6 20120102OKCDAL 2012-01-02    -8    -5     3     6
```

Plus/minus for NBA



A parallel-axis plot, by hand

The x-axis is not time. It is three scales side by side. One player is three numbers, then one line.

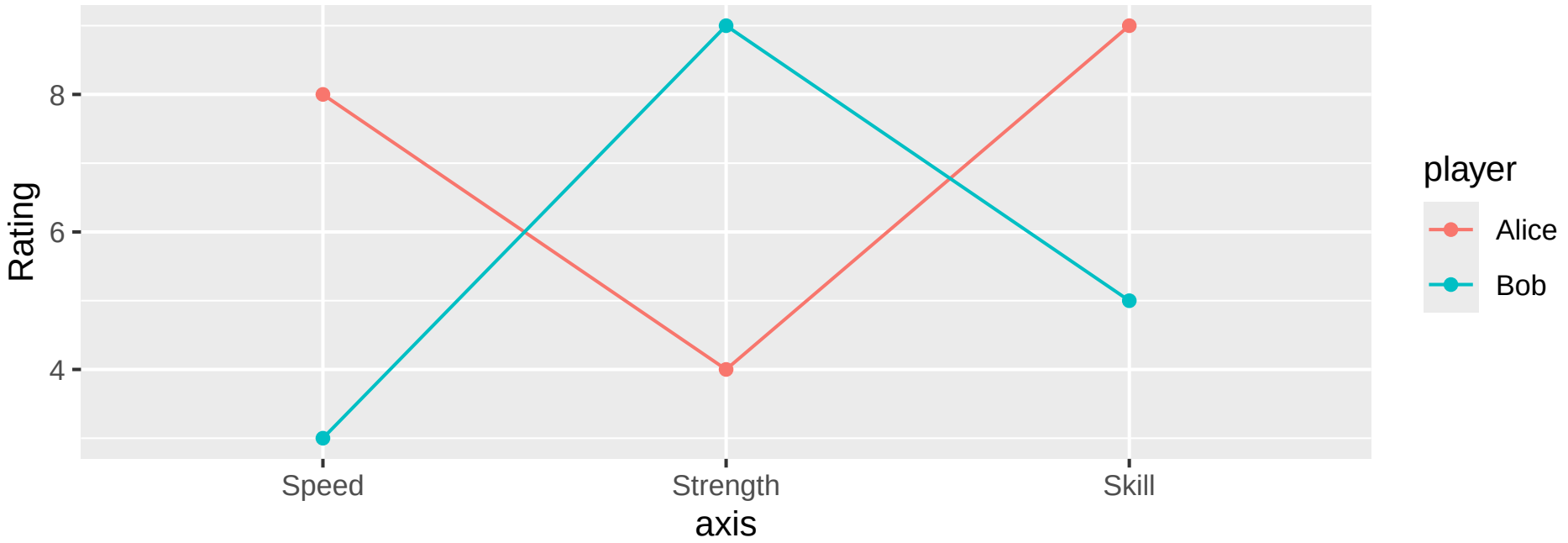
```
toy_pc <- data.frame(
  player = rep(c("Alice", "Bob"), each = 3),
  axis = factor(
    rep(c("Speed", "Strength", "Skill"), 2),
    levels = c("Speed", "Strength", "Skill")
  ),
  rating = c(8, 4, 9,  3, 9, 5)
)
toy_pc
```

```
##   player     axis rating
## 1  Alice    Speed      8
## 2  Alice Strength      4
## 3  Alice    Skill      9
## 4    Bob    Speed      3
## 5    Bob Strength      9
## 6    Bob    Skill      5
```

A parallel-axis plot, by hand

`group = player` connects the three points. Alice is high on speed and skill. Bob is high on strength.

```
ggplot(toy_pc, aes(axis, rating, group = player, color = player)) +
  geom_line() +
  geom_point() +
  ylab("Rating")
```



Roland rating, the long table

Six Thunder games, drawn at random. Same shape as the toy: one row is one game on one axis.

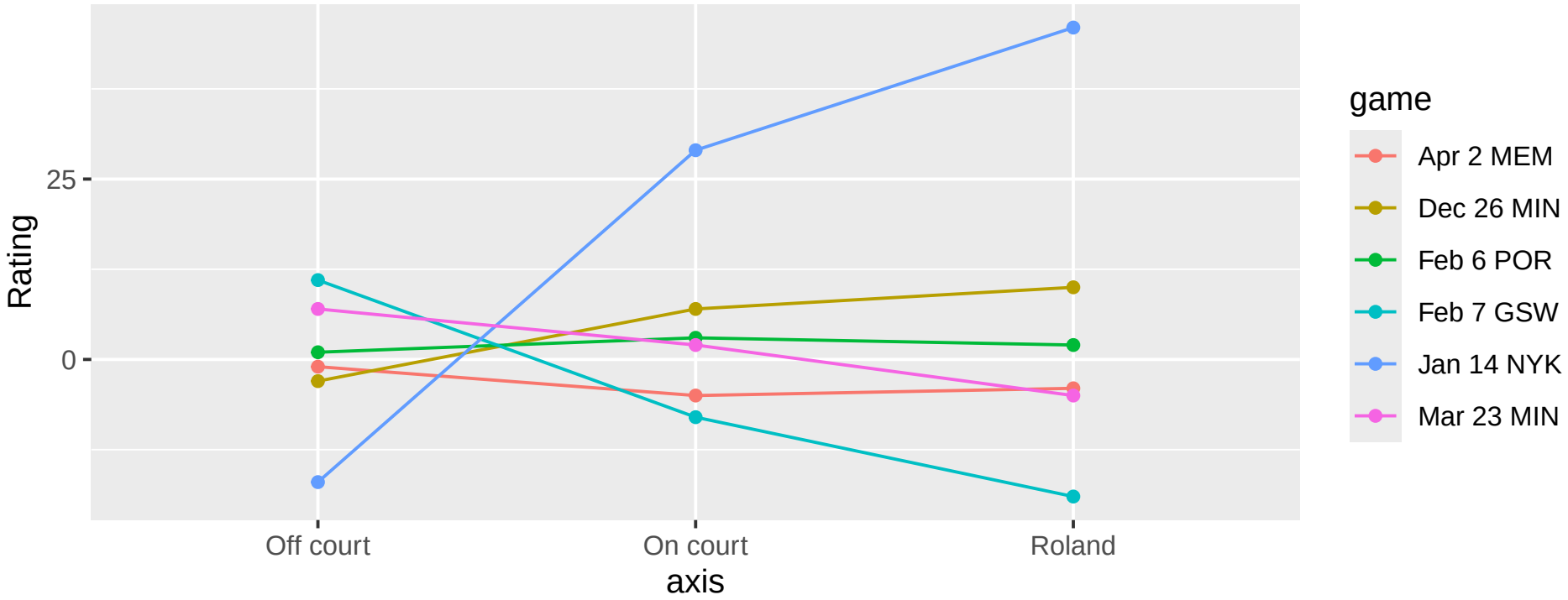
```
set.seed(270)
opp_from_id <- function(id) {
  gsub("OKC", "", sub("^[0-9]+", "", as.character(id)))
}
roland_ex <- durant_net %>%
  slice_sample(n = 6) %>%
  mutate(game = paste(md_date(Date), opp_from_id(GameID)))
pc <- roland_ex %>%
  transmute(
    game,
    `Off court` = off,
    `On court` = on,
    Roland = NET
  ) %>%
  tidyr::pivot_longer(-game, names_to = "axis", values_to = "rating") %>%
  mutate(axis = factor(axis, levels = c("Off court", "On court", "Roland")))
head(pc, 6)
```

```
## # A tibble: 6 x 3
##   game       axis      rating
##   <chr>      <fct>      <dbl>
## 1 Jan 14 NYK Off court    -17
## 2 Jan 14 NYK On court      29
## 3 Jan 14 NYK Roland        46
## 4 Feb 6 POR  Off court      1
## 5 Feb 6 POR  On court       3
## 6 Feb 6 POR  Roland         2
```

Roland rating, parallel axes

In which game did Durant have the best performance?

```
ggplot(pc, aes(axis, rating, group = game, color = game)) +
  geom_line() +
  geom_point() +
  ylab("Rating")
```



Plus/minus, on film

Raw plus/minus in a real NBA stretch. Then we adjust it for who else is on the floor.

https://www.youtube.com/watch?v=5WRe6SoBh3g

Adjusted plus/minus

Why raw plus/minus is not enough

Durant's season plus/minus credits him for every point while he is on the floor, including Westbrook's.

Adjusted plus/minus asks a different question: holding the other nine players fixed, how much does **this** player move the home margin?

Dan Rosenbaum, 82games: β_k is player k versus the reference, with the other players held constant.

The regression

One row is still one stint. The outcome is that stint's home margin. Each player is a dummy.

$$MARGIN = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

- $MARGIN = \text{PointsScoredHome} - \text{PointsScoredAway}$
- $X_k = 1$ if player k is on the **home** five
- $X_k = -1$ if player k is on the **away** five
- $X_k = 0$ if player k is **not on the floor**
- β_0 is home court. β_k is that player's adjusted plus/minus

The same stint, as a row

Home scores 2, away scores 0, so $MARGIN = +2$.

- Five home players, including Durant: $X = 1$
- Five away players: $X = -1$
- Everyone else that night, including Benson if he is sitting: $X = 0$

A season of these rows is the regression. Durant's β is his APM.

Code one stint from `nba12`

Christmas, OKC at home. Same five-name idea as `D_H`, but now with player IDs.

```
one <- nba12 %>%
  filter(GameID == "20111225ORLOKC") %>%
  slice(1)
home_ids <- as.character(c(
  one$HomePlayer1ID, one$HomePlayer2ID, one$HomePlayer3ID,
  one$HomePlayer4ID, one$HomePlayer5ID
))
away_ids <- as.character(c(
  one$AwayPlayer1ID, one$AwayPlayer2ID, one$AwayPlayer3ID,
  one$AwayPlayer4ID, one$AwayPlayer5ID
))
```

The ten IDs on the floor

Same Christmas five as before. These names built `home_ids` and `away_ids`.

Home	H1	H2	H3	H4	H5
OKC	Westbrook	Sefolosha	Durant	Ibaka	Perkins

Away	A1	A2	A3	A4	A5
ORL	Nelson	J. Richardson	Turkoglu	Anderson	Howard

```
home_ids
```

```
## [1] "786" "558" "714" "847" "253"
```

```
away_ids
```

```
## [1] "158" "90"  "151" "779" "149"
```

Durant is H3, ID 714. Nelson is A1, ID 158.

Turn an ID into 1 / -1 / 0

If the ID is in `home_ids`, code `1`. If it is in `away_ids`, code `-1`. Otherwise `0`.

```
x_code <- function(id) {
  id <- as.character(id)
  ifelse(id %in% home_ids, 1, ifelse(id %in% away_ids, -1, 0))
}
c(
  Durant = x_code(714),
  Westbrook = x_code(786),
  Nelson = x_code(158),
  Benson = x_code(1027)
)
```

```
##    Durant Westbrook    Nelson    Benson
##         1         1        -1         0
```

Benson's ID is 1027. He is in neither five, so this stint does not move his coefficient.

The APM table: nba5

That coding, for every player and every stint, is already in SportsAnalytics270. Rows are in the same order as nba12.

```
data(nba5, package = "SportsAnalytics270")
```

- 25,300 rows, 474 player columns
- Cell is 1 home, −1 away, 0 not on the floor
- Column names are player IDs. R puts an X in front of a name that starts with a digit, so 714 is X714

```
dim(nba5)
```

```
## [1] 25300   474
```

```
head(colnames(nba5), 4)
```

```
## [1] "X10"   "X1000" "X1001" "X1002"
```

One stint in nba5

Same Christmas row. nba5 should match the codes we just built from nba12.

```
i <- which(nba12$GameID == "20111225ORLOKC")[1]
nba5[i, c("X714", "X786", "X158", "X1027")]
```

```
##     X714 X786 X158 X1027
## 144    1    1   -1     0
```

Durant and Westbrook 1, Nelson −1, Benson 0. Same as x_code.

Attach the margin from `nba12`

`nba5` has the X columns only. The outcome is still home points minus away points, from `nba12`.

```
nba5 <- data.frame(
  PM = nba12$PointsScoredHome - nba12$PointsScoredAway,
  nba5
)
nba5[i, c("PM", "X714", "X786", "X158", "X1027")]
```

```
##     PM X714 X786 X158 X1027
## 144  0    1    1   -1     0
```

Christmas, 18-18: `PM = 0`. The dummies did not change. This is one row of $MARGIN = \beta_0 + \sum \beta_k X_k$.

Fit the regression

`PM ~ .` means: home margin, explained by every player column.

- `X = 1`: on the home five, this `PM` is counted as his
- `X = -1`: on the away five, this `PM` is counted against him
- `X = 0`: not on the floor, this stint does not move his β

The coefficient is his APM, holding the other players fixed.

```
model <- lm(PM ~ ., data = nba5)
coeff <- sort(coefficients(model), decreasing = TRUE)
head(names(coeff), 4)
```

```
## [1] "X1027" "X453"  "X936"  "X678"
```

The names are still IDs, `X714` not `"Durant, Kevin"`.

Put names on the coefficients

`name_coeffs` strips the `X`, then joins `PlayerID` in `nba12p`.

```
tiny <- c("(Intercept)" = unname(coeff["(Intercept)"]),
          X714 = unname(coeff["X714"]),
          X786 = unname(coeff["X786"]))
tiny
```

```
## (Intercept)        X714        X786
##  0.08848202  0.07752644  0.63468143
```

```
name_coeffs(tiny, nba12p)
```

```
##        (Intercept)      Durant, Kevin Westbrook, Russell
##         0.08848202         0.07752644         0.63468143
```

```
coeff_all <- name_coeffs(coeff, nba12p)
```

A slice of `summary(model)`

`summary(model)` has 475 rows. Print the Christmas-stint players, then the 95% range from `confint()`.

```
ids <- c("(Intercept)", "X714", "X786", "X158", "X1027")
s <- summary(model)$coefficients[ids, ]
ci <- confint(model)[ids, ]
```

Player	Estimate	SE	Low	High
home court	0.088	0.020	0.049	0.128
Durant	0.078	0.291	-0.494	0.649
Westbrook	0.635	0.326	-0.005	1.274
Nelson	0.879	0.431	0.034	1.723
Benson	2.118	1.184	-0.203	4.438

Low and High are the 95% interval: roughly Estimate $\pm 2\times$ SE.

How to read the coefficients

- **Estimate**: expected change in home `PM` if that player is on the home five (`X = 1`) instead of sitting (`X = 0`)
- **Positive** Estimate: he helps the home margin. **Negative**: he hurts it
- If he is away (`X = -1`), predicted `PM` moves by **minus** the Estimate
- **(Intercept)**: home-court baseline, about 0.09 points per stint
- **Low to High**: plausible values of that effect. If **0 is inside** the range, we cannot tell this player from sitting
- Durant 0.08, range −0.49 to 0.65: includes 0, so the point guess is not usable yet
- Benson 2.12, range −0.20 to 4.44: first on the leaderboard, but the interval is huge (about 9 minutes)
- Home court 0.05 to 0.13: does not include 0, so that effect is the one we can actually see

The first leaderboard

Do you know these players? These are β from the unfiltered model, largest first.

```
coeff_all[names(coeff_all) != "(Intercept)"][1:10]
```

```
##     Benson, Keith         Diogu, Ike   N'Diaye, Hamady    Williams, Sean
##         2.1176032         1.8681832         1.5567575         1.0007997
## Whiteside, Hassan    Nelson, Jameer     Bledsoe, Eric      Lopez, Brook
##         0.9188331         0.8786333         0.8486712         0.8390129
##     Orton, Daniel   Macklin, Vernon
##         0.8211618         0.8188564
```

What is wrong with this list?

```
nba12p %>%
  filter(PlayerTrueName == "Benson, Keith") %>%
  select(PlayerTrueName, PlayerTeams, SimpleMin)
```

```
##   PlayerTrueName PlayerTeams SimpleMin
## 1  Benson, Keith         GSW    9.1333
```

- Keith Benson played about 9 minutes for Golden State
- One lucky stint can put a player first
- With no minutes floor, the model believes a tiny sample

Rosenbaum keeps players above 250 minutes. A lower cutoff lets the lucky stint back in. A higher cutoff drops real rotation players.

Minutes in this file

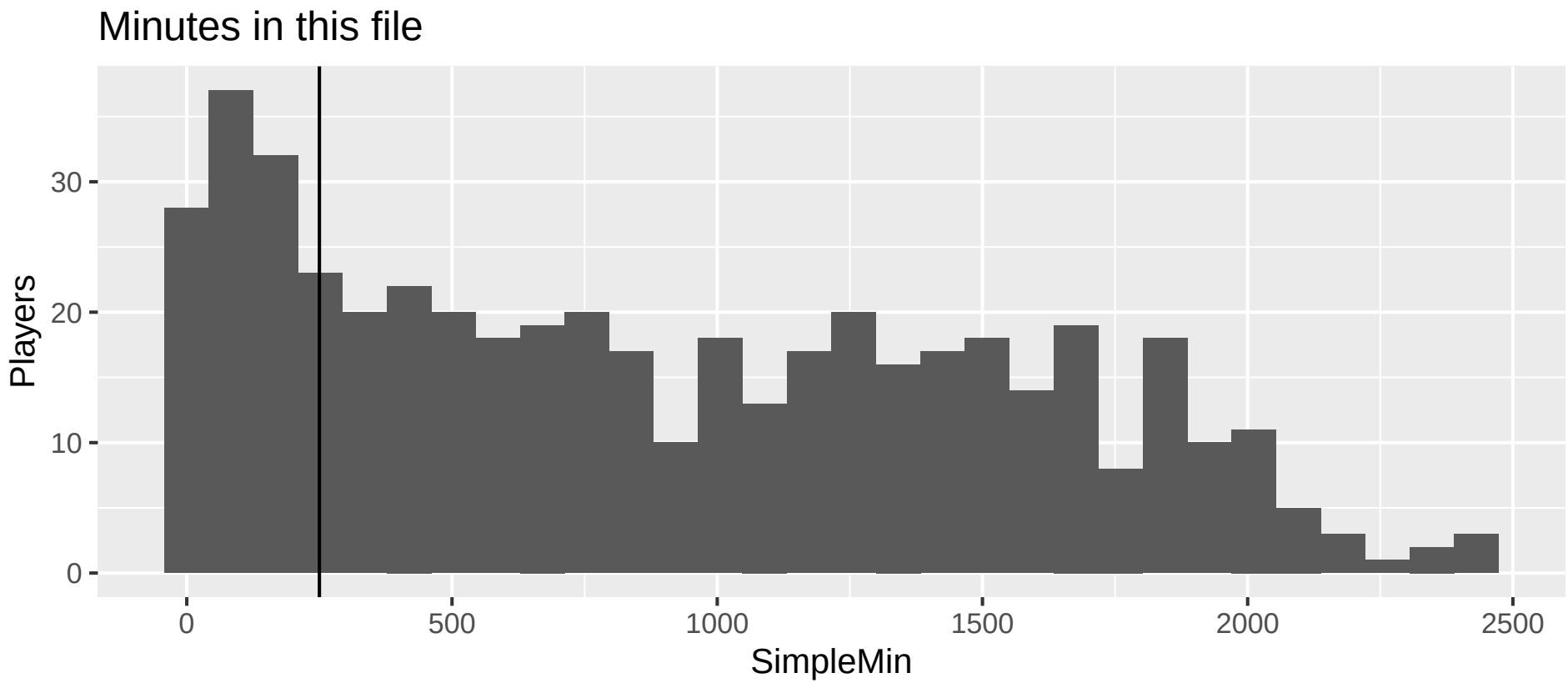
`SimpleMin` in `nba12p` is minutes in these stints. Rosenbaum keeps players above 250.

```
summary(nba12p$SimpleMin)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     0.0   292.1   812.5   892.0  1416.6  2430.9
```

Minutes in this file

Most of the mass is rotation players. The left tail is the Bensons.



Players above 250 minutes

Keep those IDs from `nba12p`. Benson (about 9 minutes) will not be here.

```
more250 <- nba12p$PlayerID[nba12p$SimpleMin > 250]
length(more250)
```

```
## [1] 372
```

```
nba12p %>%
  filter(PlayerID %in% more250) %>%
  select(PlayerID, PlayerTrueName, SimpleMin) %>%
  slice(1:4)
```

```
##   PlayerID   PlayerTrueName SimpleMin
## 1       17   Diop, DeSagana  298.3167
## 2       51 Magloire, Jamaal  315.9000
## 3      109      Ivey, Royal  352.2500
## 4      132     Okur, Mehmet  301.1000
```

```
"Benson, Keith" %in% nba12p$PlayerTrueName[nba12p$SimpleMin > 250]
```

```
## [1] FALSE
```

Strip the X from column names

nba5 columns are X714. more250 is 714. Strip the X so they match.

```
head(colnames(nba5), 4)
```

```
## [1] "PM"    "X10"   "X1000" "X1001"
```

```
colnames(nba5) <- sub("^X", "", colnames(nba5))
head(colnames(nba5), 4)
```

```
## [1] "PM"   "10"   "1000" "1001"
```

PM is unchanged. Player columns are now the same strings as PlayerID.

Keep those columns

Outcome plus the 250-minute players only. `check.names = FALSE` stops `data.frame()` from putting the `X` back on 714.

```
nba6 <- data.frame(
  PM = nba5$PM,
  nba5[, colnames(nba5) %in% as.character(more250), drop = FALSE],
  check.names = FALSE
)
c(stints = nrow(nba6), players = ncol(nba6) - 1)
```

```
##  stints players
##   25300     371
```

```
head(colnames(nba6), 5)
```

```
## [1] "PM"   "10"   "1001" "1003" "1004"
```

Refit, 250-minute players

Same `lm(PM ~ .)`, fewer columns. Nelson is no longer sitting next to Benson.

```
model2 <- lm(PM ~ ., data = nba6)
coeff2 <- sort(coefficients(model2), decreasing = TRUE)
coeff_250 <- name_coeffs(coeff2, nba12p)
coeff_250[names(coeff_250) != "(Intercept)"][1:10]
```

```
## Nelson, Jameer   Duhon, Chris Hudson, Lester    Paul, Chris   Battie, Tony
##      0.8779187      0.7791911      0.7605357      0.6760659      0.6705972
##  Bledsoe, Eric Curry, Stephen   Lopez, Robin     Wall, John Nowitzki, Dirk
##      0.6580038      0.6435656      0.6088434      0.6001202      0.5689375
```

Nelson, Jameer: https://goo.gl/eNj7sq

Where is Durant? Where is LeBron?

Rank among the 250-minute players. Still points per stint, not per possession.

```
rbind(
  apm_rank(coeff_250, "Durant, Kevin"),
  apm_rank(coeff_250, "James, LeBron")
)
```

```
##           Player        APM rank
## 1 Durant, Kevin 0.05551957  224
## 2 James, LeBron 0.48697257   15
```

Two stints are not the same length

Christmas, OKC: 18-18 in 12 and 13 possessions. A later 2-0 in one trip is a different rate. Next we divide.

$$\text{Margin} = \frac{\text{Home Points}}{\text{Home Possessions}} - \frac{\text{Away Points}}{\text{Away Possessions}}$$

Points per possession

Same scoring and possession columns as in `nba12`.

```
nba12$Margin <- nba12$PointsScoredHome / nba12$PossessionsHome -
  nba12$PointsScoredAway / nba12$PossessionsAway
nba12 %>%
  filter(GameID == "20111225ORLOKC") %>%
  select(PointsScoredHome, PointsScoredAway,
         PossessionsHome, PossessionsAway, Margin) %>%
  slice(1:3)
```

```
##   PointsScoredHome PointsScoredAway PossessionsHome PossessionsAway    Margin
## 1               18               18              12              13 0.1153846
## 2                0                0               3               2 0.0000000
## 3                0                0               1               1 0.0000000
```

First stint 18-18 on 12 and 13 trips: raw `PM` is 0, the rate is about 0.12.

Zero possessions

`Margin` is home points per possession minus away points per possession. New York has 0 home possessions, so the first fraction is 0/0.

```
nba12 %>%
  filter(GameID == "20111225BOSNYK", PossessionsHome == 0) %>%
  select(PointsScoredHome, PointsScoredAway,
         PossessionsHome, PossessionsAway, Margin) %>%
  slice(1:3)
```

```
##   PointsScoredHome PointsScoredAway PossessionsHome PossessionsAway Margin
## 1                0                2               0               1    NaN
## 2                0                0               0               1    NaN
## 3                0                0               0               1    NaN
```

In R, $0/0 =$ `NaN`. Then `NaN - 2` is still `NaN`. `lm` cannot use that row. Next slide drops it with `is.finite()`.

Drop non-finite rows

`nba6` rows still line up with `nba12`. Replace raw `PM` with the rate, then keep finite rows only.

```
nba6$PM <- nba12$Margin
nba7 <- nba6[is.finite(nba6$PM), ]
c(before = nrow(nba6), after = nrow(nba7))
```

```
## before  after
##  25300  21642
```

Secs is not a player. Add it only as a weight, later.

Per-possession regression

Same 250-minute players. `PM` is now points per possession.

```
model3 <- lm(PM ~ ., data = nba7)
coeff3 <- sort(coefficients(model3), decreasing = TRUE)
coeff_poss <- name_coeffs(coeff3, nba12p)
coeff3 <- coeff_poss
```

Per-possession leaderboard

Units changed. Nash at about 0.35 means four average teammates plus Nash score about 0.35 points per possession more than five average players.

```
coeff3[names(coeff3) != "(Intercept)"][1:10]
```

```
##     Nash, Steve  Nelson, Jameer    Mahinmi, Ian  Hudson, Lester    Duhon, Chris
##       0.3510010       0.3371446       0.3298229       0.3235370       0.2747485
##   Millsap, Paul   Kapono, Jason Humphries, Kris    Rondo, Rajon  Curry, Stephen
##       0.2464340       0.2452987       0.2439246       0.2404490       0.2359479
```

Steve Nash, Phoenix: https://goo.gl/BczeBv

Where is Durant? Where is LeBron?

Same rank helper, new units.

```
rbind(
  apm_rank(coeff3, "Durant, Kevin"),
  apm_rank(coeff3, "James, LeBron")
)
```

```
##           Player        APM rank
## 1 Durant, Kevin 0.08948282  102
## 2 James, LeBron 0.20877634   20
```

Durant and LeBron, four numbers

Same two players. The columns are not in the same units, so do not subtract them.

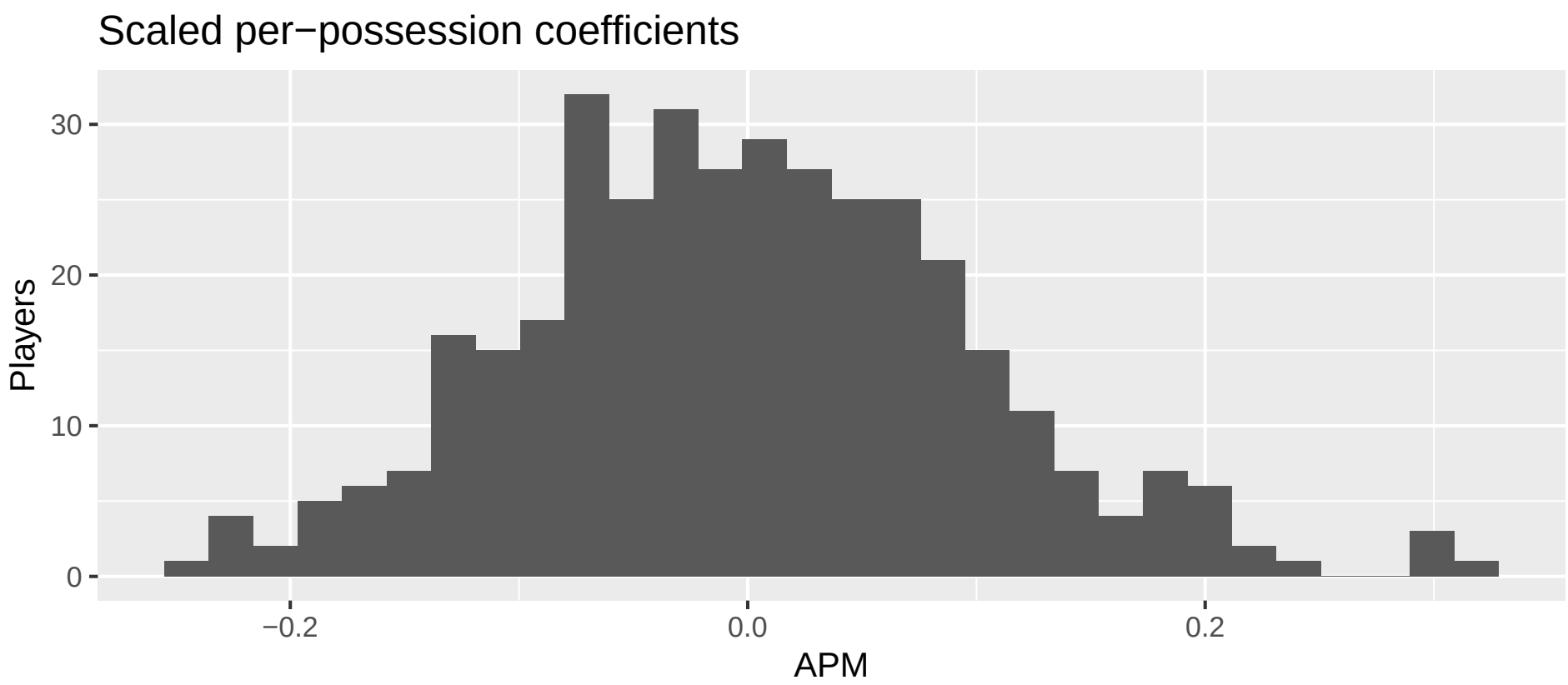
- Season plus/minus: a point total over the stints in this file
- Unfiltered and 250-minute: points per stint
- Per possession: points per possession

```
##   Player SeasonPM Unfiltered     Min250 PerPoss
## 1 Durant      324 0.07752644 0.05551957   0.089
## 2 LeBron      414 0.61816378 0.48697257   0.209
```

Center the coefficients

Subtract the mean. Then 0 is the average player in this model: positive is better than average, negative is worse.

```
coeff3 <- coeff3 - mean(coeff3)
```



Other weights you could use

- Duration: a two-minute stint should count more than six seconds
- Clutch: 4th quarter or OT, under five minutes, lead inside X points
- Garbage time: 4th quarter or OT, lead larger than X

X is often 5 or 10. Next we weight by duration only.

Weighted regression

Stint length is not equal

`ElapsedSecs` is already in `nba12`. Christmas first stint is 364 seconds. A 6-second stint is in the same file.

```
nba12 %>%
  filter(GameID == "20111225ORLOKC") %>%
  select(ElapsedSecs, PointsScoredHome, PointsScoredAway, Margin) %>%
  slice(1:3)
```

```
##   ElapsedSecs PointsScoredHome PointsScoredAway    Margin
## 1         364               18               18 0.1153846
## 2          62                0                0 0.0000000
## 3          24                0                0 0.0000000
```

Weight by minutes

`nba7` rows are the finite-`Margin` rows of `nba12`, in the same order. `Secs` is a **weight**, not a player column, so it stays out of `lm`.

```
w <- nba12$ElapsedSecs[is.finite(nba12$Margin)] / 60
c(weights = length(w), rows = nrow(nba7))
```

```
## weights    rows
##   21642   21642
```

```
model4 <- lm(PM ~ ., data = nba7, weights = w)
coeff4 <- name_coeffs(sort(coefficients(model4), decreasing = TRUE), nba12p)
coeff4[names(coeff4) != "(Intercept)"][1:10]
```

```
##    Bledsoe, Eric Magloire, Jamaal   Hudson, Lester     Lopez, Robin
##        0.2800209        0.2475193        0.2367697        0.2324336
##    Kapono, Jason   Nelson, Jameer     Duhon, Chris      Paul, Chris
##        0.2320104        0.2313027        0.2286625        0.2197614
##      Gibson, Taj   Curry, Stephen
##        0.2153007        0.2141133
```

APM is help, not a talent ranking

- APM is not a claim about how good or bad the player is as a person or as a scorer
- It is how much he moved the team's margin, given who else was on the floor

APM regressed against other statistics

Dan Rosenbaum suggests to run the following regression to understand the impact of the box score statistics on APM

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + \epsilon$$

Where Y is the coefficients of the players from Plus/Minus model,

$X_1, ..X_k$ are box score statistics

Each coefficient is the effect of that box-score statistic on APM.

APM regressed against other statistics

Basketball Statistic	Coefficient Estimate	Standard Error	Sample Mean	Standard Deviation
Pure adjusted plus/minus rating (dependent variable)	N/A		0.00	4.99
Points per 40 minutes	0.757	0.231	15.55	4.37
Effective field goal attempts per 40 minutes	-1.879	0.402	15.03	3.89
(Effective field goal attempts per 40 minutes)^2	0.026	0.010	241.1	123.3
Three point attempts per 40 minutes	0.323	0.139	2.46	2.21
Free throw attempts per 40 minutes	0.366	0.217	4.02	1.78
Assists per 40 minutes	-0.293	0.277	3.54	2.09
Offensive rebounds per 40 minutes	0.054	0.333	2.00	1.23
Defensive rebounds per 40 minutes	-0.615	0.271	4.99	1.82
Turnovers per 40 minutes	-2.171	0.480	2.36	0.65
Steals per 40 minutes	2.166	0.496	1.31	0.45
Blocks per 40 minutes	1.564	0.327	0.84	0.84
Personal fouls per 40 minutes	0.392	0.281	3.57	1.13
(Points per 40 * Assists per 40 * Rebounds per 40)$^{1/3}$	2.262	0.486	6.71	1.38
Minutes per game	0.108	0.037	27.48	8.44
Intercept	-0.043	3.083	N/A	
R^2	0.4397			
Sample size	420			

Other metrics

- Box-Score PM (Statistical Plus/Minus)- use Box-Score statistics for each stint of time of time for each player as inputs (steals, rebounds, offensive rating, etc)
- ESPN-Real Plus/Minus – is calculated by ESPN again using some sort of regression analysis.
- Offensive Plus/Minus and Defensive Plus/Minus
- Win shares: Contribution of each player into wins of the team.
- VORP – Value Over Replacement Level

Where to find the ratings?

Plus/Minus

- Best lineup by plus/minus
  https://stats.nba.com/lineups/traditional/
- ESPN: Offensive Real Plus/Minus, Defensive real Plus/Minus, RPM and Wins share (from 2014) https://www.espn.com/nba/statistics/rpm/_/sort/RPM
- Plus minus by player
  https://www.basketball-reference.com/players/d/duranke01/gamelog/2012/

Other Metrics

- WINVAL by Wayne Winston: The exact method of calculating tha Adjusted P/M is not public, however.

Margin of victory = sum(Rating of all 5 home team players) - sum(Rating of all 5 away team players)

Check

Five questions

- A stint ends 2-0 for the home team. What is the plus/minus of a home player? Of an away player?
- In the adjusted model, what does an away player get in that row: 1, -1, or 0?
- Why can a player with about 9 minutes finish first?
- Points and minutes move together. Is that skill, or time on the floor?
- Nash is about 0.35 points per possession above four average teammates. What does that claim, and what does it not claim?